

On the clustering behavior of sliding windows

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Abstract

Things can go spectacularly wrong when clustering timeseries data that has been preprocessed with a sliding window. We highlight three surprising failures that emerge depending on how the window size compares with the timeseries length. In addition to computational examples, we present theoretical explanations for each of these failure modes, and we supply a novel remedy that factors through the quotient with respect to translation.

1 Introduction

Clustering is one of the most common tasks in data science, and given the ubiquity of timeseries data, one is naturally inclined to cluster it. In order to perform Euclidean clustering (such as k -means clustering) on timeseries data, one must first map the data into Euclidean space. This is traditionally accomplished with a *sliding window*. Explicitly, given a timeseries

$$\begin{bmatrix} x(0) & x(1) & x(2) & \cdots & x(m-1) \end{bmatrix}$$

and a nonnegative integer parameter w , a sliding window of length w transforms the timeseries into $m - w + 1$ vectors in $\mathbb{R}^w$, namely, the columns of the matrix

$$\begin{bmatrix} x(0) & x(1) & x(2) & \cdots & x(m-w) \\ x(1) & x(2) & x(3) & \cdots & x(m-w+1) \\ x(2) & x(3) & x(4) & \cdots & x(m-w+2) \\ \vdots & \vdots & \vdots & & \vdots \\ x(w-1) & x(w) & x(w+1) & \cdots & x(m-1) \end{bmatrix}.$$

Naively, this seems like a reasonable approach to analyze timeseries data, but it turns out that for whatever reason, things tend to go awry with this data science pipeline. Such problems were reported by Keogh and Lin in their provocative paper “Clustering time-series sequences is meaningless” [3]. They identified that in many cases, the clustering is *meaningless* in the sense that a randomly initialized k -means algorithm returns a highly variable set of centroids.

Inspired by Keogh and Lin [3], we performed our own investigation into clustering sliding windows of timeseries data, and we were shocked by what we observed. In some cases, you can predict the centroid means from the *multiset* of timeseries values (i.e., “time” doesn’t matter). In other cases, you can predict how the windows will cluster *without even seeing the data*. Which of these behaviors emerge depends on how the window size w compares with the timeseries length m . We also demystify the emergence of sine waves in one of the experiments presented by Keogh and Lin [3]. In this paper, we provide mathematically rigorous explanations for three failure modes of raw window clustering, justifying the need for data transformations like z -normalization, and we propose a quotient embedding of z -normalized sliding windows to extract meaningful features from sliding windows.

While Keogh and Lin [3] considered both k -means clustering and hierarchical clustering, for theory reasons, we focus on k -means clustering and spectral clustering. To perform spectral clustering, first run principal component analysis to find the most representative $(k-1)$ -dimensional affine subspace, and then project the data onto that affine subspace before solving k -means for the projected data.

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It turns out that spectral clustering is a 2-approximation algorithm for k -means [2], and empirically, spectral clustering-optimal centroids are very similar to the k -means-optimal centroids.

Throughout, we 0-index our timeseries, and we denote $[p] := \{0, \dots, p-1\}$. Also, timeseries have length m , windows have size w , $n = m - w + 1$ denotes the total number of windows, and k denotes the desired number of clusters.

2 Small windows have flat cluster centroids

The technology company Apple went public on December 12, 1980 under the ticker symbol AAPL, and today on Kaggle, one can access the closing price of this stock for every trading day over the subsequent 42 years [4]. In light of the exponential growth in this stock, we take a logarithm to obtain the timeseries of length 10590 shown in Figure 1.

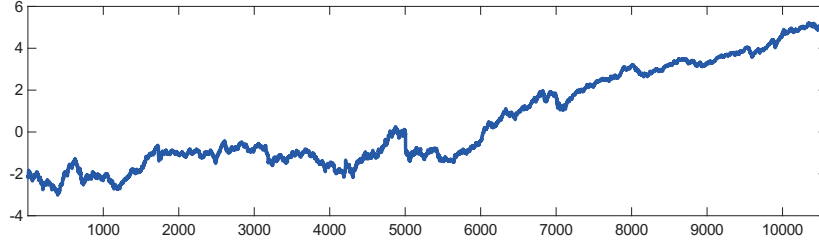


Figure 1: The log AAPL time series.

Running k -means clustering with $k = 10$ on all windows of length 100 from this timeseries then results in the centroids shown in Figure 2.

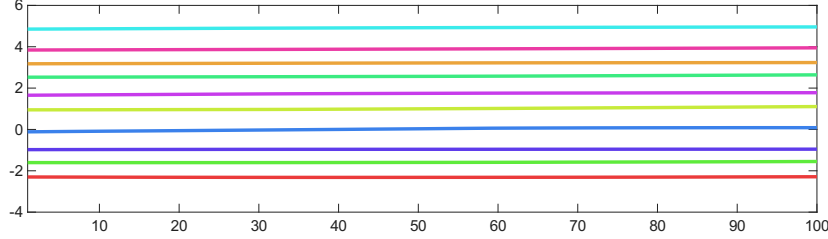


Figure 2: Centroids of raw sliding windows from the log AAPL time series

Every centroid is nearly constant! Instead of illustrating typical fluctuations in the timeseries over the course of 100 timesteps, these centroids essentially capture how the individual timeseries values cluster together. In this section, we show that this “flat centroid” phenomenon emerges as a result of the window length being “too small” relative to the full timeseries length. For theory reasons, we focus on spectral clustering with only $k = 2$ clusters.

As is often the case in timeseries analysis, zero padding will serve as a convenient preprocessing step in what follows. To be explicit, the $(w-1)$ -zero-padding of x is defined to be the timeseries $\tilde{x}$ obtained by appending $w-1$ zeros to both sides of x :

t	0	1	$\dots$	$w-2$	$w-1$	w	$\dots$	$m+w-2$	$m+w-1$	$m+w$	$\dots$	$m+2w-3$
$\tilde{x}(t)$	0	0	$\dots$	0	$x(0)$	$x(1)$	$\dots$	$x(m-1)$	0	0	$\dots$	0

Notably, the left-most w -window of $\tilde{x}$ contains $x(0)$, while the right-most window contains $x(m-1)$. We denote $\text{pad} := \{0, 1, \dots, w-2\} \cup \{m+w-1, m+w, \dots, m+2w-3\}$.

Theorem 1. *Given a timeseries x of length m , let $\mathcal{X}$ denote the multiset of all w -windows of the $(w-1)$ -zero-padding of x . Then the following hold:*

- (a) *The centroid μ of $\mathcal{X}$ is a scalar multiple of the all-ones vector $\mathbf{1}$.*
- (b) *For $k = 2$, the spectral clustering-optimal centroids μ_1 and μ_2 of $\mathcal{X}$ both satisfy*

$$\frac{\|P_{1^\perp} \mu_i\|_2^2}{\|\mu_i - \mu\|_2^2} \leq \frac{8w^2 \|x\|_\infty^2 + w^3 m \|x\|_{\text{Lip}}^2}{\|x - \bar{x}\mathbf{1}\|_2^2},$$

64 where $P_{1^\perp}$ is the orthogonal projection onto the orthogonal complement of the span of 1 , $\bar{x}$ denotes
 65 the average coordinate of x , and $\|x\|_{\text{Lip}} := \max_{t \in [m-1]} |x(t+1) - x(t)|$.

66 Part (a) above is a strengthening of Theorem 1 in [3] that holds in our special case. Before proving
 67 Theorem 1, we evaluate part (b) with a couple of standard examples of x . If x is a random walk, then
 68 we expect $\|x\|_\infty \asymp \sqrt{m}$, $\|x\|_{\text{Lip}} = 1$, and $\|x - \bar{x}1\|_2 \asymp \sqrt{m}$, and so the right-hand side in (b) above is on
 69 the order of $\frac{w^3}{m}$, which is small when $w \ll m^{1/3}$. Meanwhile, if $x = \sin \frac{2\pi \cdot}{p}$ for some period $p \gg 1$, then
 70 $\|x\|_\infty \asymp 1$, $\|x\|_{\text{Lip}} \asymp \frac{1}{p}$, and $\|x - \bar{x}1\|_2 \asymp \sqrt{m}$, and so our error estimate is on the order of $\frac{w^2}{m} + \frac{w^3}{p^2}$,
 71 which is small when $w \ll \min\{m^{1/2}, p^{2/3}\}$.

72 *Proof of Theorem 1.* For (a), since $\mathcal{X}$ consists of all w -windows of the $(w-1)$ -zero-padding of x , each
 73 coordinate of μ is the average of the m coordinates of x along with $w-1$ zeros. For (b), note that
 74 μ_1 and μ_2 both reside in the 1-dimensional affine subspace delivered by principal component analysis.
 75 Accordingly, let X denote the matrix of column vectors from $\mathcal{X}$, center these column vectors by taking
 76 $B := X - \mu 1^\top$, and let u denote the top left-singular vector of B . Since there necessarily exist $c_1, c_2 \in \mathbb{R}$
 77 such that $\mu_i = \mu + c_i u$ for each i , it follows that

$$78 \frac{\|P_{1^\perp} \mu_i\|_2^2}{\|\mu_i - \mu\|_2^2} = \|P_{1^\perp} u\|_2^2 = 1 - \langle u, \frac{1}{\sqrt{w}} 1 \rangle^2 \leq \frac{\|B - \frac{1}{w} 1 1^\top B\|_F^2}{\sigma_1(B)^2} = \frac{\|P_{1^\perp} X\|_F^2}{\sigma_1(B)^2},$$

79 where the inequality follows from Wedin's $\sin \Theta$ theorem [7]. We estimate the denominator by

$$80 \sigma_1(B)^2 \geq \|e_1^\top B\|_2^2 \geq \|x - \bar{\mu} 1\|_2^2 \geq \|x - \bar{x} 1\|_2^2.$$

81 Next, we estimate the numerator in terms of the $(w-1)$ -zero-padding $\tilde{x}$ of x :

$$82 \|P_{1^\perp} X\|_F^2 = \sum_{s=0}^{m+w-2} \sum_{t=0}^{w-1} \left(\tilde{x}(s+t) - \frac{1}{w} \sum_{t'=0}^{w-1} \tilde{x}(s+t') \right)^2 \leq \sum_{s=0}^{m+w-2} \sum_{t=0}^{w-1} \left(\frac{1}{w} \sum_{t'=0}^{w-1} |\tilde{x}(s+t) - \tilde{x}(s+t')| \right)^2.$$

83 Considering

$$84 |\tilde{x}(s+t) - \tilde{x}(s+t')| \leq \begin{cases} 2\|x\|_\infty & \text{if } s \in \text{pad or } s+w-1 \in \text{pad} \\ (w-1)\|x\|_{\text{Lip}} & \text{else,} \end{cases}$$

85 the result follows. $\square$

86 3 Clustering nearly symmetric data is meaningless

87 In his PhD thesis [6], Naoki Saito generated a synthetic dataset to evaluate various classification
 88 algorithms. This dataset, which is available online [5], consists of independent draws from three
 89 specially crafted distributions of random timeseries of length 128, and each distribution is named after
 90 the shape of the resulting timeseries: *cylinder*, *bell*, and *funnel*; see Example 4.7 in [6] for details.
 91 Running k -means on the raw windows (which consists of 90 windows with 30 examples of each type)
 92 results in the centroids shown in Figure 3.

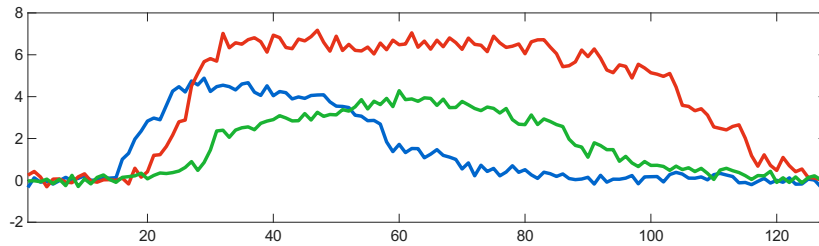


Figure 3: Centroids of the cylinder–bell–funnel dataset.

93 Let us use the cylinder–bell–funnel dataset to replicate an experiment described in Section 4 of [3].
 94 Concatenate the examples in the training set to form a timeseries of length $90 \cdot 128 = 11520$, and then

run k -means clustering with $k = 3$ on all windows of length 128 from this timeseries. The resulting centroids are depicted in Figure 4.

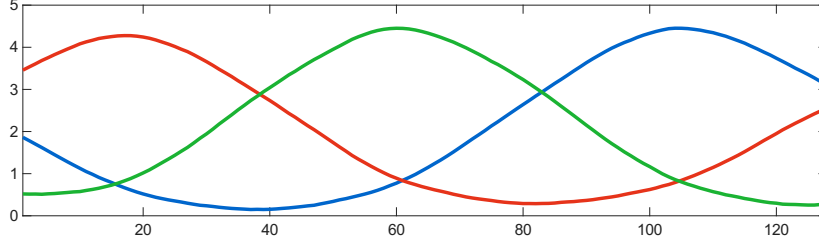


Figure 4: Centroids of sliding windows from the cylinder–bell–funnel timeseries.

Every centroid is nearly a sine wave! This comes as a surprise since none of the examples in the training set exhibit such regularity. However, the discrete Fourier transform of our 11520-long timeseries offers a hint for what’s going on here: The dominant sinusoidal contribution has period 128 (i.e., w), with all others being much smaller by comparison. In this section, we show that (nearly) sinusoidal centroids emerge in such settings.

The following result is written in terms of a certain notion of distance between subspaces. The *chordal distance* between two r -dimensional subspaces of $\mathbb{R}^d$ is given by

$$\sqrt{\sin^2 \theta_1 + \cdots + \sin^2 \theta_r},$$

where $\theta_1, \dots, \theta_r$ denote the principal angles between these subspaces. This defines a metric on the Grassmannian space $\text{Gr}(r, \mathbb{R}^d)$. Note that this distance is small only if all of the principal angles are small, meaning every unit vector in one of the subspaces is close to the other subspace.

Theorem 2. *Given positive integers w and p , consider the timeseries x of length $m := pw + w - 1$ defined by*

$$x(t) = a + b \sin \frac{2\pi(t-c)}{w} + e(t)$$

for some $a, b, c \in \mathbb{R}$ and some timeseries e with mean zero. Let $\mathcal{X}$ denote the multiset of all w -windows of x . Then the following hold:

- (a) The centroid μ of $\mathcal{X}$ satisfies $\|\mu\|_{\text{Lip}} \leq \frac{2}{pw} \|e\|_{\infty}$.
- (b) For $k = 3$, the spectral clustering-optimal centroids μ_1, μ_2 , and μ_3 of $\mathcal{X}$ all have the property that $\mu_i - \mu$ resides in a 2-dimensional subspace whose chordal distance from the span of $\cos \frac{2\pi \cdot}{w}$ and $\sin \frac{2\pi \cdot}{w}$ is at most

$$\left(\frac{|b|}{\sqrt{\frac{4}{pw} \|e\|_2^2 + \frac{12}{p} \|e\|_{\infty}^2}} - 1 \right)^{-1}$$

whenever this quantity is positive.

Part (a) above is yet another version of Theorem 1 in [3] that holds in our special case. Note that the bound in part (b) can be interpreted as $(\text{snr} - 1)^{-1}$, where snr is one way to quantify the *signal-to-noise ratio* in our setup. By Theorem 2, when snr is larger, we can expect the spectral clustering-optimal centroids to be more sinusoidal.

Before proving Theorem 2, we highlight the significance of our choice $m := pw + w - 1$. Suppose $p = 2$, $w = 3$, and $e(t) = 0$ for all t . Then $m = 8$, and x takes the form

$$\begin{bmatrix} \alpha & \beta & \gamma & \alpha & \beta & \gamma & \alpha & \beta \end{bmatrix}.$$

The length- w windows of x are then given by the columns of the matrix

$$\left[\begin{array}{ccc|ccc} \alpha & \beta & \gamma & \alpha & \beta & \gamma \\ \beta & \gamma & \alpha & \beta & \gamma & \alpha \\ \gamma & \alpha & \beta & \gamma & \alpha & \beta \end{array} \right].$$

In general, if x is w -periodic with length m , then we end up with pw windows that can be partitioned into p equal batches of size w , with each batch consisting of all cyclic permutations of the first window. We will exploit this structure to demonstrate the emergence of sinusoidal centroids.

Proof of Theorem 2. For (a), we have

$$\mu(t) = a - \frac{1}{pw} \sum_s e(s),$$

where the sum is over $s \in \{0, \dots, t-1\} \cup \{m-w+t+1, \dots, m-1\}$. (Here and throughout, we make use of identities like $\sum_{t=0}^{w-1} \sin \frac{2\pi(t-c)}{w} = 0$, which can be deduced by interpreting the left-hand side as the imaginary part of a vanishing geometric sum.) Thus,

$$|\mu(t+1) - \mu(t)| = \frac{1}{pw} |e(m-w+t+1) - e(t)| \leq \frac{2}{pw} \|e\|_\infty.$$

For (b), we use the fact that each μ_i resides in the 2-dimensional affine subspace given by principal component analysis. Consider the matrix X whose column vectors belong to $\mathcal{X}$, and center these vectors to get $B := X - \mu 1^\top$. It will be useful to consider the matrices A and E defined by

$$A_{s,t} = b \sin \frac{2\pi(s+t-c)}{w}, \quad E_{s,t} = a + e(s+t) - \mu(s),$$

since this allows for the decomposition $B = A + E$, where A is “structured” and E is “small”. In particular, $AA^\top$ is a circulant matrix:

$$(AA^\top)_{s,s'} = \sum_{t=0}^{pw-1} b^2 \sin \frac{2\pi(s+t-c)}{w} \sin \frac{2\pi(s'+t-c)}{w} = \frac{b^2 pw}{2} \cos \frac{2\pi(s-s')}{w}.$$

Thus, the eigenvalues of $AA^\top$ are given by the discrete Fourier transform of $\frac{b^2 pw}{2} \cos \frac{2\pi \cdot}{w}$, from which one may conclude that $AA^\top$ is $\frac{b^2 pw^2}{4}$ times the orthogonal projection onto the span of $\cos \frac{2\pi \cdot}{w}$ and $\sin \frac{2\pi \cdot}{w}$. Next, we show that E is small:

$$|E_{s,t}| = |a + e(s+t) - \mu(s)| \leq |e(s+t)| + |a - \mu(s)| \leq |e(s+t)| + \frac{1}{p} \|e\|_\infty,$$

which implies

$$|E_{s,t}|^2 \leq |e(s+t)|^2 + 2 \cdot |e(s+t)| \cdot \frac{1}{p} \|e\|_\infty + \frac{1}{p^2} \|e\|_\infty^2 \leq |e(s+t)|^2 + \frac{3}{p} \|e\|_\infty^2,$$

and so

$$\|E\|_F^2 \leq w \|e\|_2^2 + 3w^2 \|e\|_\infty^2.$$

Thus, Wedin’s $\sin \Theta$ theorem [7] and Weyl’s inequality for singular values gives that the desired chordal distance is at most

$$\frac{\|B - A\|_F}{\sigma_2(B)} \leq \frac{\|E\|_F}{\sigma_2(A) - \|E\|_F} = \left(\frac{\sigma_2(A)}{\|E\|_F} - 1 \right)^{-1} \leq \left(\frac{|b|}{\sqrt{\frac{4}{pw} \|e\|_2^2 + \frac{12}{p} \|e\|_\infty^2}} - 1 \right)^{-1}. \quad \square$$

Keogh and Lin [3] used the cylinder–bell–funnel example to illustrate their observation that clustering sliding windows is meaningless. In retrospect, the primary reason for meaninglessness in this case is the underlying symmetry: These sliding windows can be viewed as noisy versions of a single orbit under the action of cyclic permutations. Meanwhile, if we were to cluster the orbit directly, any centroids that emerge would have the same k -means value as their cyclic permutations. (We will encounter this symmetry again in Figure 9 and the surrounding discussion.) This is an instance of a more general phenomenon:

Clustering nearly symmetric data is meaningless.

For example, run k -means clustering on thousands of independent realizations of a spherical Gaussian random vector in $\mathbb{R}^2$. Since the underlying distribution is rotation invariant, we can expect any rotation of the k -means-optimal centroids to perform similarly well as cluster centroids.

4 Large windows have interval clusters

Empirically, we observe that when the window is a sizable fraction of the timeseries, the optimal clusters are frequently intervals, i.e., each cluster consists of sliding windows that are all adjacent in time. As an example, we performed 1000 trials of the following experiment: Draw a random walk of length $m = 100$, and then for each $w \in \{1, 2, \dots, m - k + 1\}$, run k -means with $k = 3$ on all w -windows of the random walk, and record whether all of the resulting clusters are intervals. See Figure 5 for a plot of the proportion of trials that produced interval clusters for each w .

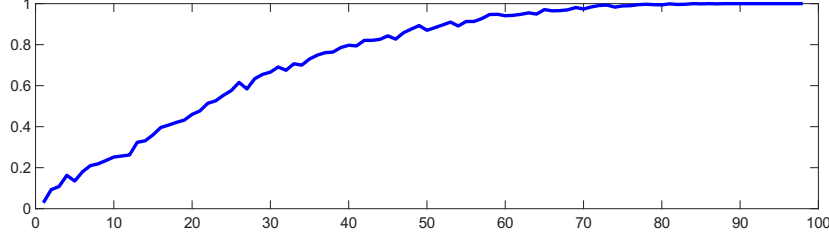


Figure 5: Empirical frequency of interval clusters versus window length.

In this section, we provide some theoretical explanation for this phenomenon.

Consider independent random variables $Z_1, \dots, Z_{m-1}$ with zero mean and unit variance, and define a random length- m timeseries x by

$$x(t) = \sum_{i=1}^t Z_i.$$

(Notably, $x(0) = 0$.) Take $n := m - w + 1$, and for each $s \in [n]$, let x_s denote the window

$$\begin{bmatrix} x(s) & x(s+1) & \dots & x(s+w-1) \end{bmatrix}.$$

We are interested in the partition $C_1 \sqcup \dots \sqcup C_k = [n]$ that minimizes the k -means objective:

$$\sum_{i=1}^k \frac{1}{|C_i|} \sum_{s, s' \in C_i} \|x_s - x_{s'}\|^2.$$

When w is large, we expect the k -means objective of each partition to concentrate at its expectation (especially when the Z_i 's have subgaussian distribution), and so the quantity to minimize becomes a slight perturbation of

$$\sum_{i=1}^k \frac{1}{|C_i|} \sum_{s, s' \in C_i} \mathbb{E} \|x_s - x_{s'}\|^2.$$

This is the setting of our main result in this section:

Theorem 3. *For the random $\{x_s\}_{s \in [n]}$ defined above, the partitions $C_1 \sqcup \dots \sqcup C_k = [n]$ that minimize*

$$\sum_{i=1}^k \frac{1}{|C_i|} \sum_{s, s' \in C_i} \mathbb{E} \|x_s - x_{s'}\|^2$$

are precisely those for which each C_i is an interval of length $\lfloor \frac{n}{k} \rfloor$ or $\lceil \frac{n}{k} \rceil$.

Before proving this theorem, we present a couple of helpful lemmas.

Lemma 4. $\mathbb{E} \|x_s - x_{s'}\|^2 = w|s - s'|$.

Proof. Without loss of generality, we have $s' < s$, in which case

$$\begin{aligned} \mathbb{E} \|x_s - x_{s'}\|^2 &= \mathbb{E} \sum_{t=0}^{w-1} \left(x(s+t) - x(s'+t) \right)^2 = \mathbb{E} \sum_{t=0}^{w-1} \left(\sum_{i=s'+t+1}^{s+t} Z_i \right)^2 \\ &= \sum_{t=0}^{w-1} \sum_{i=s'+t+1}^{s+t} \sum_{j=s'+t+1}^{s+t} \mathbb{E} Z_i Z_j = w(s - s'), \end{aligned}$$

and the result follows. $\square$

195 **Lemma 5.** For every subset $C \subseteq \mathbb{Z}$ of size r , it holds that

$$196 \quad \sum_{s, s' \in C} |s - s'| \geq \frac{r(r+1)(r-1)}{3},$$

197 with equality precisely when C is an interval.

198 *Proof.* Take $C = \{s_0, \dots, s_{r-1}\}$ with $s_0 < \dots < s_{r-1}$. Then $|s_i - s_j| \geq |i - j|$ for all $i, j \in [r]$, and so

$$199 \quad \sum_{s, s' \in C} |s - s'| = \sum_{i, j \in [r]} |s_i - s_j| \geq \sum_{i, j \in [r]} |i - j|,$$

200 with equality precisely when C is an interval. Standard summation identities simplify this lower bound:

$$201 \quad \sum_{i, j \in [r]} |i - j| = 2 \sum_{i=1}^{r-1} \sum_{j=1}^i j = \sum_{i=1}^{r-1} (i^2 + i) = \frac{r(r-1)(2r-1)}{6} + \frac{r(r-1)}{2} = \frac{r(r+1)(r-1)}{3}. \quad \square$$

202 *Proof of Theorem 3.* By Lemma 4, we seek to minimize

$$203 \quad \sum_{i=1}^k \frac{1}{|C_i|} \sum_{s, s' \in C_i} |s - s'|,$$

204 and Lemma 5 implies that the minimizers have the property that each C_i is an interval. It remains to
 205 establish that for each i , the size r_i of C_i is $\lfloor \frac{n}{k} \rfloor$ or $\lceil \frac{n}{k} \rceil$. By Lemma 5, the quantity to minimize is

$$206 \quad \sum_{i=1}^k \frac{1}{r_i} \cdot \frac{r_i(r_i+1)(r_i-1)}{3} = \frac{1}{3} \left(\sum_{i=1}^k r_i^2 - k \right)$$

207 subject to the constraints that each r_i is a positive integer and $r_1 + \dots + r_k = n$.

208 If $r_i > r_j + 1$, then it's better to take $r_i \leftarrow r_i - 1$ and $r_j \leftarrow r_j + 1$:

$$209 \quad (r_i - 1)^2 + (r_j + 1)^2 = r_i^2 + r_j^2 - 2(r_i - r_j - 1) < r_i^2 + r_j^2.$$

210 As such, $(r_1, \dots, r_k)$ is suboptimal if any two coordinates differ by more than 1. The only remaining
 211 options take each r_i to be $\lfloor \frac{n}{k} \rfloor$ or $\lceil \frac{n}{k} \rceil$. Furthermore, the constraint $r_1 + \dots + r_k = n$ forces $n \bmod k$ of
 212 the coordinates to equal $\lceil \frac{n}{k} \rceil$ and the other coordinates to equal $\lfloor \frac{n}{k} \rfloor$. All of these remaining choices
 213 have the same value, so they all must be optimal. $\square$

214 5 Data transformations accentuate clustering structure

215 In settings where data fails to exhibit clustering structure, a data scientist is inclined to apply a
 216 transformation (e.g., a feature map) that accentuates the desired clustering structure. In the context
 217 of timeseries data, the go-to transformation is *z-normalization*, where for each window, the entry
 218 values are translated to have zero mean and then scaled to have unit variance. Such a transformation
 219 is particularly relevant in settings where the window mean and variance are independent of the desired
 220 signal. For example, in hindsight, the result of our first experiment on “small windows have flat cluster
 221 centroids” was overly influenced by the window means. As an alternative, applying *z-normalization* in
 222 the same experiment results in the following centroids, which are decidedly *not* flat; see Figure 6.

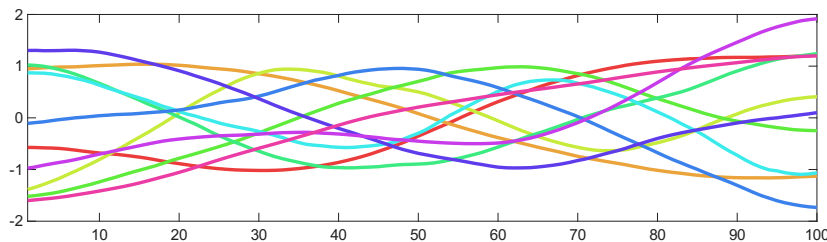


Figure 6: Centroids of *z*-normalized sliding windows from the log AAPL time series.

223 Sadly, these centroids resemble the “meaningless” sinusoids that plagued our second “clustering
 224 nearly symmetric data is meaningless” example. In fact, z -normalization also fails to improve the
 225 situation there; see Figure 7.

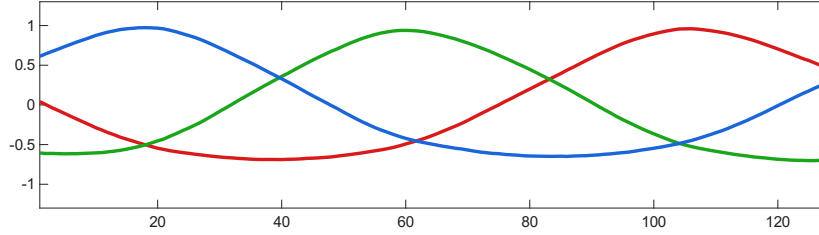


Figure 7: The $k = 3$ centroids after z -normalization are shown in red, green, and blue, which are all of sinusoidal shapes with some phase shifts. Compared with the raw window clustering centroids in Figure 4, the main difference is the vertical scale of the plotted centroids, since z -normalization enforces zero mean and unit variance for each window.

226 Finally, z -normalization delivers better results in our third experiment on “large windows have
 227 interval clusters”; see Figure 8.

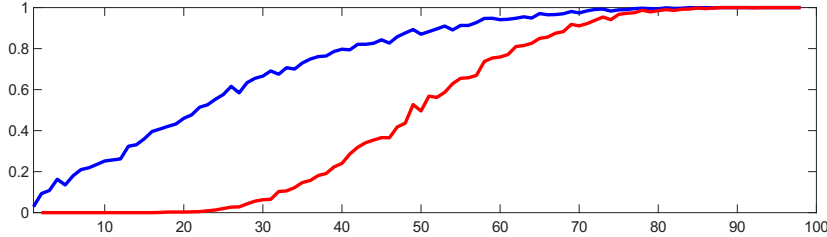


Figure 8: Empirical frequency of interval clusters for raw windows versus window length (in blue) compared to the same for z -normalized windows (in red).

228 In particular, while it is still the case that large windows tend to be optimally clustered into
 229 intervals, the effect does not persist as much to smaller windows after z -normalization. To summarize,
 230 we observe three different effects of z -normalization in our three experiments:

- 231 1. z -normalization prevents flat centroids.
- 232 2. z -normalization does not prevent sinusoidal centroids.
- 233 3. z -normalization prevents interval clusters when the windows are not too large.

234 Since z -normalization only improves the situation in our third experiment, we are left wondering what
 235 data transformation could possibly help with our first two experiments. There has been some work
 236 on *motifs* [8] and *matrix profiles* [9] that are designed to extract the desired clustering structure from
 237 sliding windows, but we had difficulty making these ideas work in our settings. Instead, we conclude
 238 this section by illustrating what appears to be a new (and intuitive!) approach for accentuating the
 239 clustering structure of sliding windows.

240 We start by illustrating the sliding windows from our second experiment, which arise from a time-
 241 series generated from the cylinder–bell–funnel dataset. See Figure 9(left) for a visualization of the
 242 z -normalized sliding windows. Notably, the sliding windows form what appear to be a regular sample
 243 of a continuous circular curve. Every sliding window is a concatenation of portions from two different
 244 “pure windows,” which in turn are plotted in large dots. After inspecting the geometry of this dataset,
 245 it is perhaps not surprising that the k -means cluster centroids fail to recover the planted cylinder, bell,
 246 and funnel shapes.

247 Much like k -means clustering, the multidimensional embedding in Figure 9(left) is fundamentally
 248 based on the *Euclidean distance* between windows, but perhaps a different choice of distance would
 249 carry more semantic meaning. To guide our intuition, consider a *sparse* timeseries, meaning most of

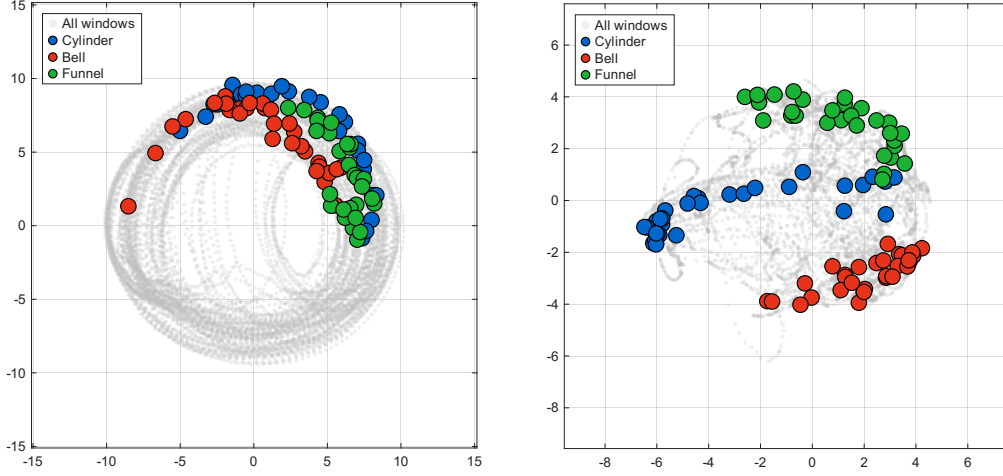


Figure 9: Multidimensional scaling embedding of z -normalized sliding windows under Euclidean distance (left) and under the C_w -invariant pseudometric d (right).

the entries are zero, and let us take sliding windows *without* z -normalization. When we take sliding windows of such a timeseries, neighboring windows will frequently have disjoint support. For example, here are two neighboring windows of length $w = 8$ from a sparse timeseries:

$$x_0 := (0, 0, 0, 2, 0, 0, -2, 0) \quad \text{and} \quad x_1 := (0, 0, 2, 0, 0, -2, 0, 0).$$

Since these windows have disjoint support, they are extremely far from each other in Euclidean distance. But should we not think of these windows as being close to one another? As an alternative, consider the cyclic group C_w acting by circular translation on the space $\mathbb{R}^w$ of windows. This suggests a different notion of distance between points in $\mathbb{R}^w$:

$$d(x, y) := \min_{s \in [w]} \left(\sum_{i \in [w]} (x(i+s) - y(i))^2 \right)^{1/2},$$

where $i+s$ is interpreted modulo w . Notably, in our example above, x_0 and x_1 can be viewed as circular translations of each other, and so $d(x_0, x_1) = 0$, which better aligns with our intuition. Of course, since $x_0 \neq x_1$, it follows that d is not quite a metric on $\mathbb{R}^w$. Instead, d is a C_w -invariant *pseudometric* that descends to a metric on the orbit space $\mathbb{R}^w/C_w$. As such, this pseudometric implicitly implements a different type of data transformation: it maps a window to its orbit under the action of C_w , i.e., $\mathbb{R}^w \rightarrow \mathbb{R}^w/C_w$. By passing to these orbits, we approximately mod out by the (non-circular) translation structure that underlies sliding windows. The effect on our cylinder–bell–funnel dataset is nicely illustrated in Figure 9(right): the sliding windows no longer trace out a “meaningless” circular path in the embedding space, and the “pure windows” now exhibit the separation needed for clustering.

There is another useful phenomenon that emerges when we pass to the orbit space. While the sliding windows still form a curve that jumps from one “pure window” to another, more of the curve is spent near the “pure windows.” In fact, sliding windows p' that have substantial overlap with a “pure window” p have the property that $d(p', q) \approx d(p, q)$, where q denotes any other window. This is most striking when we plot $d(x, q)$ over sliding windows x for a fixed window q as in Figure 10: the quotient distances consistently exhibit small variation near “pure windows.”

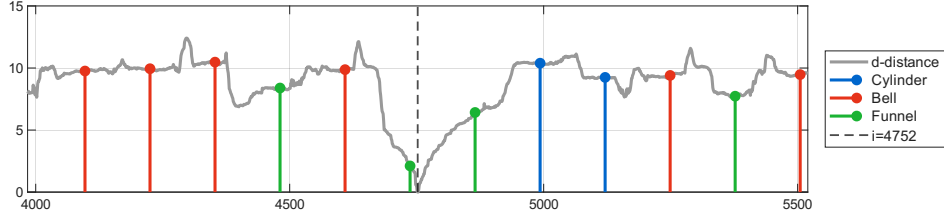


Figure 10: The quotient distance from a fixed reference window 4752 to nearby windows in the original time order. The colored stems denote “pure windows.”

In order to obtain a meaningful clustering, it behooves us to discard any unhelpful “mixed windows.” We can accomplish this by (1) filtering out points with uncommon quotient distances, and (2) only keeping the points whose quotient distances have small local variation. Here’s a description of an iterative *filtered k -medoids algorithm* that implements this idea:

1. **Initialization.** Run the k -means++ initialization under the quotient distance d to select k candidate windows.
2. **Filter for common quotient distances.** For each candidate window, collect the sliding windows that exhibit the most common quotient distances, using several bin widths for robustness.
3. **Filter for small local variation in quotient distance.** Select a further subset of M windows whose quotient distances to the current k candidates have the smallest local variation.
4. **Cluster the survivors.** Partition the surviving M windows according to the nearest candidate in quotient distance.
5. **Update the candidates.** Replace each candidate with the medoid of its cluster, i.e., the cluster member with the smallest average quotient distance to the other members.
6. **Termination.** Stop if the medoids change by less than certain tolerance, or if the number of iterations exceeds a certain tolerance. Otherwise, return to step 2.

Steps 4 and 5 mirror the update steps of Lloyd’s algorithm, except the centers are updated by medoids rather than centroids. See the GitHub repository [1] for the details of our implementation. Applying this algorithm to the cylinder–bell–funnel timeseries with $k = 3$ and $M = 90$ results in the (meaningful!) medoids given in Figure 11.

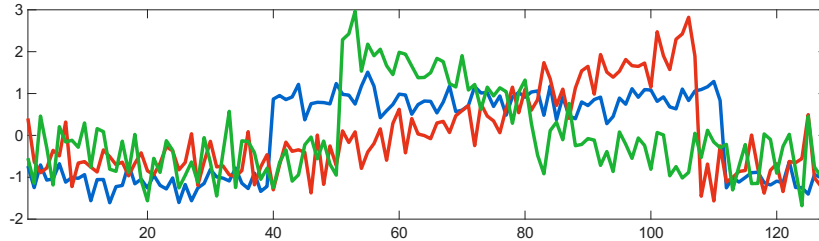


Figure 11: Medoids of the cylinder–bell–funnel timeseries, produced by the iterative filtered k -medoids algorithm. As one might hope, the medoids resemble “pure windows,” one of each shape.

Returning to the log AAPL timeseries, we also run our algorithm with $k = 10$ and $M = 600$ to obtain the decidedly non-sinusoidal medoids displayed in Figure 12.

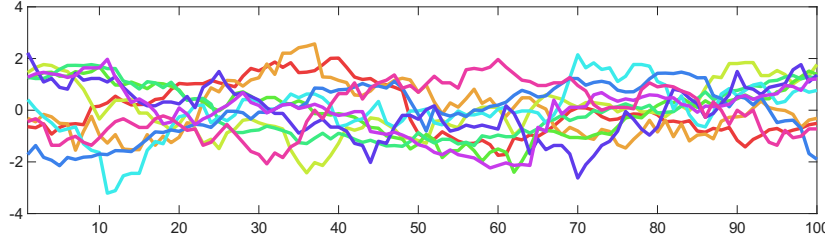


Figure 12: Medoids of the log AAPL time series, produced by the iterative filtered k -medoids algorithm.

While this approach appears to avoid the “meaningless” traps that arise when clustering sliding windows in Euclidean space, it also presents its own difficulties. Most notably, we observe that different random seeds lead to medoids that look essentially the same but differ by a cyclic shift. (This is an artifact of the cyclic invariance of our pseudometric.) Also, our iteration is not guaranteed to converge thanks to the filtering steps. We leave these issues for future work.

6 Discussion

In this paper, we highlighted a few phenomena that arise when clustering sliding windows. We now point out two observations that warrant further investigation.

First, Theorems 1 and 2 are both restrictive in how to select k . In the setting of Theorem 1, we empirically observe flat cluster centroids when $k > 2$, and in the setting of Theorem 2, we similarly observe sinusoidal clusters when $k > 3$. However, to prove both theorems, we argue by relating the covariance matrix of the data to another matrix of some rank r , and so we can only prove guarantees for spectral clustering with $k \leq r + 1$. Of course, if the rank of the covariance matrix is *exactly* r , then the optimal clustering centroids necessarily reside in the r -dimensional affine subspace implicated by principal component analysis, even if $k > r + 1$. Is there a way to estimate the proximity of $k > r + 1$ optimal clustering centroids to this affine subspace in general?

Second, we pointed out that clustering nearly symmetric data is meaningless, but it would be interesting to determine the extent to which this explains the meaninglessness observed by Keogh and Lin [3]. In particular, is it always the case that when sliding windows exhibit meaningless clustering, there exists an approximate symmetry that acts on the windows? And is this approximate symmetry necessarily given by cyclic permutations?

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